



St Catherine's School
Waverley

2016

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Task Weighting – 40%

Total Marks – 100

Section I Pages 3 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section.

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1. Evaluate $\frac{2.48 \times 0.034}{\sqrt{0.081} - 0.029}$, giving your answer correct to 2 decimal places.

(A) 0.32

(B) 0.33

(C) 0.36

(D) 0.37

2. Solve for x , $|4x + 2| = 6$

(A) $x = 1$, $x = -2$

(B) $x = -1$, $x = 2$

(C) $x = 1$, $x = 2$

(D) $x = -1$, $x = -2$

3. Express 215° in radian measure.

(A) $\frac{\pi}{215}$

(B) $\frac{215}{\pi}$

(C) $\frac{43\pi}{36}$

(D) $\frac{36\pi}{43}$

4. Which of the following is equal to $\frac{\cos(\frac{\pi}{2} - \alpha)}{\sin(2\pi - \alpha)}$?

(A) 1

(B) $\cot \alpha$

(C) -1

(D) $\cos(\frac{\pi}{2})$

5. Find $\int (2x + 1)^5 dx$

(A) $\frac{(2x+1)^6}{6} + c$

(B) $5(2x + 1)^4 + c$

(C) $10(2x + 1)^4 + c$

(D) $\frac{(2x+1)^6}{12} + c$

6. $\frac{3\sqrt{3}}{\sqrt{7}-2}$ is equal to:

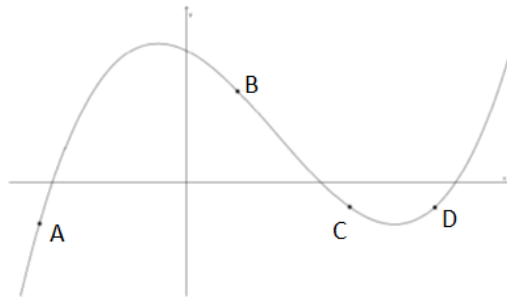
(A) $\frac{3\sqrt{3}(\sqrt{7}+2)}{5}$

(B) $\frac{3\sqrt{3}(\sqrt{7}-2)}{5}$

(C) $\sqrt{3}(\sqrt{7} + 2)$

(D) $\sqrt{3}(\sqrt{7} - 2)$

7. Which point on the graph satisfies the description: $y < 0$, $\frac{dy}{dx} > 0$, $\frac{d^2y}{dx^2} < 0$



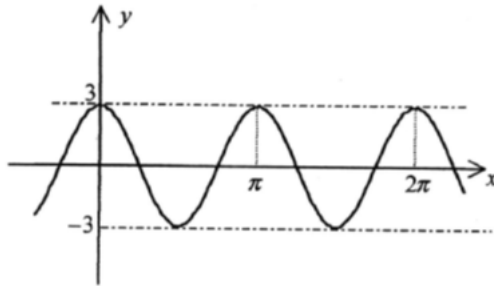
(A) Point A

(B) Point B

(C) Point C

(D) Point D

8. The graph below represents $y = a \cos mx$. Which statement is correct?



- (A) $a = -3, m = 1$ (B) $a = 3, m = 2$
(C) $a = 3, m = 1$ (D) $a = 1, m = 3$
9. If $\log_a 7 = x$ and $\log_a 3 = y$, evaluate $\log_a 63$.
- (A) $x + y^2$ (B) $x + 2y$
(C) xy^2 (D) $2x + y$
10. The common difference d of the arithmetic series $\ln 8 + \ln 16 + \ln 32 + \dots$ is:
- (A) 1 (B) 2
(C) $\ln 2$ (D) $\ln 8$

END OF SECTION I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) If $\sqrt{50} - 3\sqrt{75} + \sqrt{18} = a\sqrt{2} - b\sqrt{3}$, find the values of a and b . 2
- (b) State the domain and range of the function $y = \sqrt{2x - 1} - 4$ 2
- (c) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x}$ 2
- (d) Differentiate with respect to t : 2
 $y = (5t^4 - 8)^9$
- (e) Find a primitive function of $5x^3 + \sin 4x$ 2
- (f) A function $y = f(x)$ has $\frac{d^2y}{dx^2} = 6x - 2$ and a stationary point at $(3, 0)$. 3
Find $f(x)$.
- (g) For the parabola $x^2 - 4x - 8y - 4 = 0$, find the coordinates of the vertex. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Solve the equation for x : 3

$$3 \times 9^x + 2 \times 3^x - 1 = 0$$

- (b) α and β are the roots of the quadratic equation $3x^2 - 4x - 8 = 0$. Without calculating the roots, find the value of:

(i) $\alpha^2 + \beta^2$ 2

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ 1

- (c) Differentiate the following with respect to x :

(i) $y = xe^{\sin x}$ 2

(ii) $y = \frac{\ln x}{x}$ 2

- (d) (i) Find $\int \frac{x^3+1}{x^2} dx$ 2

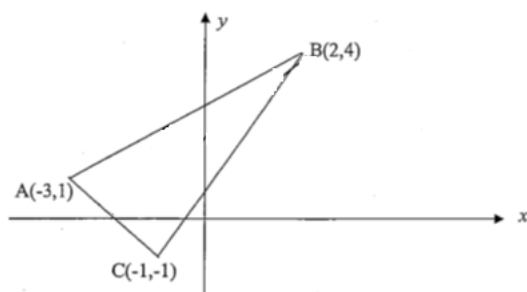
(ii) Show that $\frac{d}{dx} \left(\sin x - \frac{1}{3} \sin^3 x \right) = \cos^3 x$ 2

Hence, find $\int 3\cos^3 x \, dx$ 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a)



In the diagram above $A(-3,1)$, $B(2,4)$ and $C(-1,-1)$.

- | | |
|--|---|
| (i) Show that $\triangle ABC$ is isosceles. | 2 |
| (ii) Find the equation of AC. | 2 |
| (iii) Find the perpendicular distance of point B from the line AC. | 1 |
| (iv) Find the area of $\triangle ABC$. | 1 |

(b) Consider the function $y = 5xe^{-x}$.

- | | |
|--|---|
| (i) Copy and complete the table below, giving the value correct to 2 decimal places. | 1 |
|--|---|

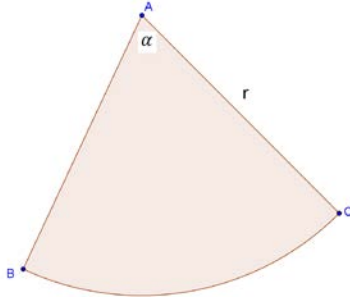
x	0	1	2	3	4
y	0			0.75	0.37

- | | |
|--|---|
| (ii) Using Simpson's rule with 5 function values evaluate $\int_0^4 5xe^{-x} dx$ | 2 |
| (iii) Show that $\frac{d^2y}{dx^2} = 5e^{-x}(x - 2)$. | 2 |
| (iv) Find the stationary points and the points of inflection. | 4 |

End of Question 13

Question 14 (15 marks) Use a **SEPARATE** writing booklet

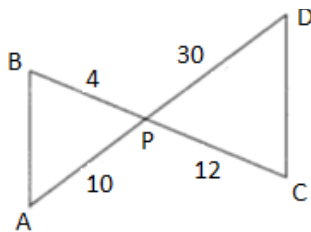
(a)



The diagram above shows a sector with radius r and angle α . The area of the sector is $625m^2$.

- | | | |
|-------|--|----------|
| (i) | Find the expression for α in terms of r . | 1 |
| (ii) | Show that the perimeter of the sector is $P = 2r + \frac{1250}{r}$ | 1 |
| (iii) | Find the value of r such that the sector has minimum perimeter. | 3 |
| (iv) | Find the value of α to the nearest degree such that the perimeter is minimum. | 1 |

b)



In the diagram above, AD and BC intersect at point P so that AP=10, BP=4, CP=12 and DP=30.

- | | | |
|------|--|----------|
| (i) | Prove that $\triangle ABP$ is similar to $\triangle CPD$. | 2 |
| (ii) | Hence, prove that $AB \parallel CD$. | 2 |

Question 14 continues on page 10

The population P of a town is changing exponentially $P = P_0 e^{kt}$, t is the time in years.

(c)

- (i) At the start of 2000, the population was 15000 people and at the start of 2010 it was 35000. Show that the growth rate k is approximately 8.5%. **3**
- (ii) Using $k = 0.085$, after how many years the population will reach 50000 people? **2**

End of Question 14

Question 15 (15 marks) Use a **SEPARATE** writing booklet

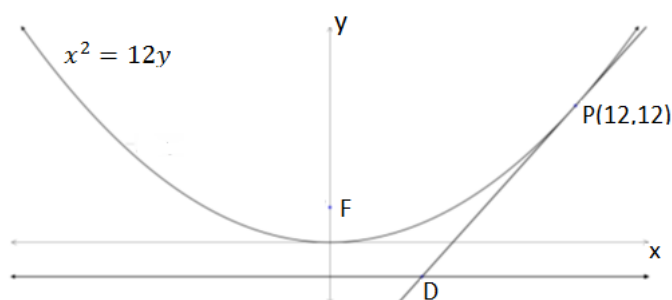
- (a) The velocity of a particle is given by $v = \frac{4}{2t+1} \text{ ms}^{-1}$. Initially the particle is 2m to the left of the origin.

- (i) Find the expression for the displacement x . 2
- (ii) Find the exact time when the particle is at the origin. 2
- (iii) Show that the acceleration of the particle is always negative. 2

- (b) Two cards are chosen at random without replacement from six cards numbered: 1, 1, 3, 4, 4, 4. What is the probability that:

- (i) First card is 1 and second card is 4? 1
- (ii) The sum of two numbers on the cards is less than 5? 2

(c)



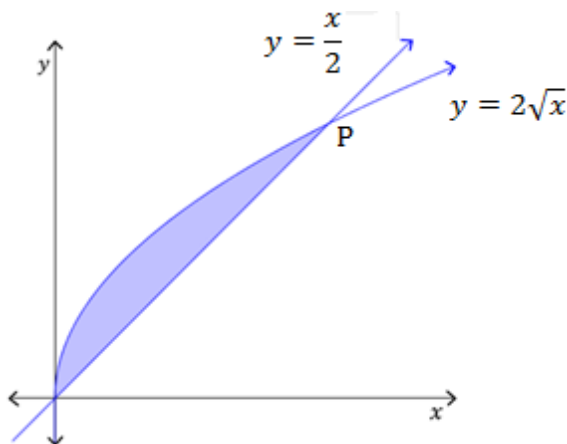
$P(12,12)$ is a point on the parabola $x^2 = 12y$. F is the focus of the parabola. The tangent of the parabola at P intersects the directrix at point D .

- (i) Find the equation of the tangent at point P . 2
- (ii) Show that the coordinates of point D are $(\frac{9}{2}, -3)$. 2
- (iii) Show that $\angle PFD = 90^\circ$. 2

End of Question 15

Question 16 (15 marks) Use a **SEPARATE** writing booklet

- (a) The region between the curve $y = 2\sqrt{x}$ and $y = \frac{x}{2}$ is rotated about the x axis.



- (i) Show that the coordinates of point P are 8 and 16. **1**
- (ii) Find the volume of the solid in exact form. **2**
- (b) Kate wants to save \$30000 for her holiday. She invests \$150 at the beginning of each month. Interest is paid at the rate of 12% per annum compounded monthly.
- (i) How much money will Kate save after 5 years? **2**
- (ii) How many months will it take Kate to reach her goal? **2**
- (c) Consider the function $y = 1 - 2 \sin x$.
- (i) Find the exact values where the function y cuts the x-axis for $0 \leq x \leq 2\pi$. **2**
- (ii) Find the values of y when $x = 0$ and $x = 2\pi$. **1**
- (iii) Draw a neat sketch of the function y for $0 \leq x \leq 2\pi$. **2**
- (iv) Find the area bounded by the curve $y = 1 - 2 \sin x$, the x-axis and the lines $x = 0$ and $x = \frac{\pi}{2}$, in exact form. **3**

End of paper

Student Number: Solutions

2016 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics

Multiple Choice Answer Sheet

Completely fill the response circle representing the most correct answer

	A	B	C	D
1.	☐	☒	☐	☐
2.	☒	☐	☐	☐
3.	☐	☐	☒	☐
4.	☐	☐	☒	☐
5.	☐	☐	☐	☒
6.	☐	☐	☒	☐
7.	☒	☐	☐	☐
8.	☐	☒	☐	☐
9.	☐	☒	☐	☐
10.	☐	☐	☒	☐

2016 Maths Trial (solutions)

① 0.33 B

② $|4x+2| = 6$

$$4x+2=6$$

$$4x=4$$

$$x=1$$

$$-4x+2=-6$$

$$4x=-8$$

$$x=-2$$

A

③ 215°

$$180^\circ = \pi$$

$$215^\circ = \frac{\pi}{180} \times 215$$

$$= \frac{43\pi}{36}$$

C

④ $\frac{\cos(\frac{\pi}{2}-\alpha)}{\sin(2\pi-\alpha)} = \frac{\sin \alpha}{-\sin \alpha} = -1$ C

⑤ $\int (2x+1)^5 dx = \frac{(2x+1)^6}{6 \times 2} + C$ D

$$= \frac{(2x+1)^6}{12} + C$$

⑥ $\frac{3\sqrt{3}}{\sqrt{7}-2} \times \frac{\sqrt{7}+2}{\sqrt{7}+2} = \frac{3\sqrt{3}(\sqrt{7}+2)}{7-4} = \sqrt{3}(\sqrt{7}+2)$ C

⑦ A

⑧ B

⑨ $\log_a 7 = x$ $\log_a 3 = y$

$$\log_a 63 = \log_a 7 \times 9 = \log_a 7 + \log_a 3^2 = \log_a 7 + 2\log_a 3$$
$$= x + 2y$$

B

⑩ $\ln 8 + \ln 16 + \ln 32 + \dots$
 $\ln 2^3 + \ln 2^4 + \ln 2^5 + \dots$
 $3\ln 2 + 4\ln 2 + 5\ln 2 + \dots$ $\therefore d = \ln 2$ C

Question 11

(11)

a) $\sqrt{50} - 3\sqrt{75} + \sqrt{18} = a\sqrt{2} - b\sqrt{3}$

LHS: $5\sqrt{2} - 15\sqrt{3} + 3\sqrt{2}$

$= 8\sqrt{2} - 15\sqrt{3}$

$\therefore a=8 \quad b=15$

- 1 mark for SIMPLIFYING CORRECTLY
- 2 marks for CORRECT VALUES OF a AND b .

b) $y = \sqrt{2x-1} - 4$

D: $2x-1 \geq 0$

$2x \geq 1$

$x \geq \frac{1}{2}$

R: $\sqrt{2x-1} \geq 0$

$\sqrt{2x-1} - 4 \geq -4$

domain: all real $x \geq \frac{1}{2}$

range: all real $y \geq -4$

- 1 mark for CORRECT DOMAIN OR RANGE
- 2 marks for CORRECT DOMAIN AND RANGE

c) $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x} = \lim_{x \rightarrow 0} \frac{1}{4} \left(\frac{\sin 5x}{5x} \right)^{\rightarrow 1} \times 5 = \frac{5}{4}$

- 1 mark for EXPRESS LIMIT AS A FUNCTION OF $\lim_{x \rightarrow 0} \frac{\sin 5x}{5x}$
- 2 marks for CORRECT ANSWER

d) $y = (5t^4 - 8)^9$

$y' = 9(5t^4 - 8)^8 \times 20t^3$

$= 180t^3(5t^4 - 8)^8$

- 1 mark for SUBSTANTIAL EFFORT TO DIFFERENTIATE CORRECTLY
- 2 marks for CORRECT ANSWER

e) $\int (5x^3 + \sin 4x) dx = 5x \frac{x^4}{4} - \frac{\cos 4x}{4} + c$
 $= \frac{5x^4}{4} - \frac{\cos 4x}{4} + c$

- 1 mark for INTEGRATING A TERM CORRECTLY
- 2 marks for: CORRECT ANSWER

Question 11 (cont.)

f) $y'' = 6x - 2$ S.P. at $(3, 0)$

$$y' = \int (6x - 2) dx$$

$$y' = \frac{6x^2}{2} - 2x + c$$

$$= 3x^2 - 2x + c$$

S.P. at $(3, 0) \therefore y' = 0$

$$27 - 6 + c = 0 \therefore c = -21$$

$$y' = 3x^2 - 2x - 21$$

$$y = \int (3x^2 - 2x - 21) dx$$

$$y = 3 \frac{x^3}{3} - \frac{2x^2}{2} - 21x + c_1$$

$$y = x^3 - x^2 - 21x + c_1$$

$x = 3, y = 0$

$$27 - 9 - 63 + c_1 = 0 \therefore c_1 = 45$$

$$f(x) = x^3 - x^2 - 21x + 45$$

- 1 mark for CORRECT PRIMITIVE OF y''
- 2 marks for CORRECT PRIMITIVE OF y'
- 3 marks for CORRECT ANSWER

g)(i) $x^2 - 4x - 8y - 4 = 0$

$$x^2 - 4x = 8y + 4$$

$$x^2 - 4x + 4 = 8y + 4 + 4$$

$$(x - 2)^2 = 8(y + 1)$$

$$\vee (2, -1)$$

- 1 mark for PUTTING EQUATION IN THE FORM $(x - h)^2 = 4a(y - k)$ OR SUBSTANTIAL EFFORT USING A CORRECT TECHNIQUE TO FIND THE VERTEX
- 2 marks for CORRECT ANSWER

Question 12

a) $3 \times 9^x + 2 \times 3^x - 1 = 0$
 $3 \times (3^x)^2 + 2 \times 3^x - 1 = 0$

let $3^x = t$

$$3t^2 + 2t - 1 = 0$$

$$t = \frac{-2 \pm \sqrt{4 + 12}}{6} = \frac{-2 \pm 4}{6}$$

$$t = -1, t = \frac{1}{3}$$

$$t = -1 \quad \text{or} \quad t = \frac{1}{3}$$

$$3^x = -1$$

no solutions

$$3^x = \frac{1}{3}$$

$$3^x = 3^{-1} \therefore \boxed{x = -1}$$

- 1 mark for correct simplification to a quadratic equation
- 2 marks for correct solving of the quadratic equation
- 3 marks for correct solution of x and dismissing one solution

b)

$$3x^2 - 4x - 8 = 0$$

(i) $\alpha + \beta = -\frac{b}{a}$

$$\alpha\beta = \frac{c}{a}$$

$$\alpha + \beta = \frac{4}{3}$$

$$\alpha\beta = -\frac{8}{3}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \frac{16}{9} + \frac{16}{3}$$

$$= \frac{16 + 48}{9}$$

$$= \frac{64}{9}$$

- 1 mark for correct sum and product
- 2 marks for recognising $(\alpha + \beta)^2 - 2\alpha\beta$ and correct answer

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{\frac{64}{9}}{-\frac{8}{3}} = -\frac{8}{3}$

- 1 mark for correct answer

Question 12 (cont.)

c) (i) $y = x e^{\sin x}$

$$y' = e^{\sin x} + x e^{\sin x} \times \cos x \\ = e^{\sin x} (1 + x \cos x)$$

- 1 mark for correctly using the product rule
- 2 marks for correct answer

(ii) $y = \frac{\ln x}{x}$

$$y' = \frac{\frac{1}{x} \times x - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

- 1 mark for correctly using either the quotient or product rule
- 2 marks for correct answer

d)

(i) $\int \frac{x^3 + 1}{x^2} dx = \int \frac{x^3}{x^2} dx + \int \frac{dx}{x^2}$
 $= \int x dx + \int x^{-2} dx$

$$= \frac{x^2}{2} + \frac{x^{-1}}{-1} + C$$

$$= \frac{x^2}{2} - \frac{1}{x} + C$$

- 1 mark for splitting and simplifying the fraction
- 2 marks for correct integration + C

ii) $\frac{d}{dx} \left(\sin x - \frac{1}{3} \sin^3 x \right) = \cos x - \frac{3}{3} \sin^2 x \cos x$

$$= \cos x (1 - \sin^2 x)$$

$$= \cos x \times \cos^2 x$$

$$= \cos^3 x$$

- 1 mark for correct differentiation
- 2 marks for substituting trig. identity to show the RHS.

Question 12 (cont.)

$$\int 3 \cos^3 x \, dx = 3 \int \cos^3 x \, dx$$

$$= 3 \left(\sin x - \frac{1}{3} \sin^3 x \right) + C$$

$$= 3 \sin x - \sin^3 x + C$$

- 1 mark for correct integration

Question 13

a) A (-3, 1) B (2, 4) C (-1, -1)

i) $d(A, B) = \sqrt{5^2 + 3^2} = \sqrt{34}$

$d(A, C) = \sqrt{4^2 + 2^2} = \sqrt{20}$

$d(B, C) = \sqrt{9 + 25} = \sqrt{34}$

$AB = BC \therefore \triangle ABC$ is isosceles.

- 1 mark for 1 correct distance
- 2 marks for 2 correct dist. and making a statement

(ii) AC: $m_{AC} = \frac{2}{-2} = -1$

$$y + 1 = -1(x + 1)$$

$$y + 1 = -x - 1$$

$$x + y + 2 = 0$$

- 1 mark for gradient
- 2 marks for correct application of $y - y_1 = m(x - x_1)$

(iii) B (2, 4) $x + y + 2 = 0$

$$d = \frac{|2 + 4 + 2|}{\sqrt{2}} = \frac{8}{\sqrt{2}} \quad \text{or} \quad \frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

- 1 mark for correct subst. and simplification

Question 13 (cont.)

$$(iv) A = \frac{1}{2} AC \times d = \frac{1}{2} \times \frac{\sqrt{20}}{\sqrt{2}} \times 4\sqrt{2} = \frac{2\sqrt{40}}{\sqrt{2}} = 4\sqrt{10} \text{ u}^2$$

• 1 mark for correct application of area

b)

	x_0	x_1	x_2	x_3	x_4
(i)	0	1	2	3	4
y	0	1.84	1.35	0.75	0.37

• 1 mark for y value

$$(ii) \int_a^b f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4 \times \text{odds} + 2 \times \text{evens}]$$

$$h = \frac{4-0}{4} = 1$$

$$\int_0^4 5xe^{-x} dx = \frac{1}{3} [0 + 0.37 + 4(1.84 + 0.75) + 2 \times 1.35] = 4.48$$

- 1 mark for h and substitution
- 2 marks for correct application and answer

$$(iii) y' = 5e^{-x} + 5xe^{-x} \times (-1) = 5e^{-x} - 5xe^{-x}$$

$$y'' = -5e^{-x} - 5e^{-x} + 5xe^{-x} = 5e^{-x}(x-2)$$

- 1 mark for y'
- 2 marks for y''

Question 13 (cont.)

(iv) $y' = 5e^{-x}(1-x)$

$y' = 0$ if $1-x=0 \therefore x=1$

$1-x$	+	-	$1-x > 0$	$e^{-x} > 0$
y'	+	-	$x < 1$	
y	↗	↘		

for $x=1$ $y=1.84$

$(1, 1.84)$ is a max

$y'' = 5e^{-x}(x-2)$

$y'' = 0$ if $x-2=0 \therefore x=2$

$(2, 1.35)$ is a possible point of inflection

$5e^{-x} > 0$

$x-2$	-	+
y''	-	+
y	∩	∪

$x-2 > 0$

$x > 2$

concavity changes

$(2, 1.35)$ is a point of inflection

- 1 mark for x-coord. of SP.
- 2 marks for SP
- 3 marks for SP and a point of inflexion
- 4 marks for testing the point of inflex: + all above

Question 14

Q) i) $A = \frac{1}{2} r^2 \alpha$

$$625 = \frac{1}{2} r^2 \alpha \quad \therefore \alpha = \frac{2 \times 625}{r^2}$$

$$\alpha = \frac{1250}{r^2}$$

• 1 mark for $\alpha = \frac{1250}{r^2}$

(ii) $P = r + r + r\alpha$
 $= 2r + \frac{1250}{r}$

• 1 mark for correct answer

(iii) $P = 2r + 1250 r^{-1}$

$$\frac{dP}{dr} = 2 + 1250 r^{-2} \times (-1)$$

$$= 2 - \frac{1250}{r^2}$$

$$= \frac{2r^2 - 1250}{r^2}$$

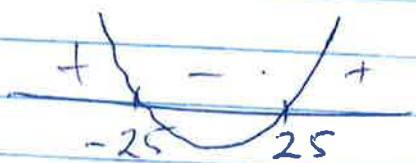
$$\frac{dP}{dr} = 0 \quad \text{if} \quad 2r^2 - 1250 = 0$$

$$r^2 - 625 = 0$$

$$r^2 = 625$$

$$r = 25$$

$$(r-25)(r+25) = 0$$



	25	
$\frac{dP}{dr}$	-	+
P	↘	↗

at 25 $\rightarrow$ min.

• 1 mark for $2r^2 - 1250 = 0$

• 2 marks for $r = 25$

• 3 marks for checking $r = 25$ is min.

Question 14 (cont.)

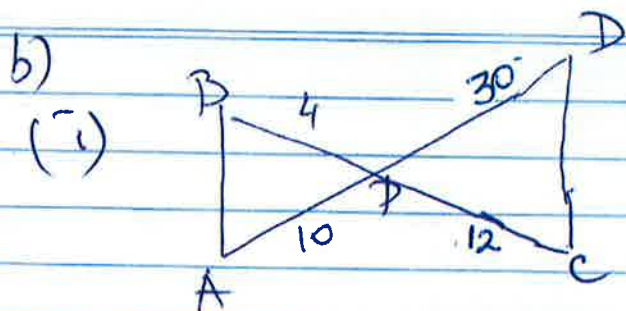
(iv) for $r=25$

$$\alpha = \frac{1250}{25^2} = 2 \text{ rad.}$$

$$\alpha \approx 115^\circ$$

$$\pi = 180^\circ$$
$$2 \text{ rad} = \frac{180^\circ}{\pi} \times 2$$

• 1 mark for α in degrees



$\triangle APB$ and $\triangle CPD$

$\angle APB = \angle CPD$ (vertically opposite angles are equal.)

$$\frac{AP}{PB} = \frac{10}{4} = \frac{5}{2}$$

$$\frac{PD}{PC} = \frac{30}{12} = \frac{5}{2}$$

2 pairs of matching sides are in the same ratio and included angles are equal

$\therefore \triangle APB \parallel \triangle CPD$

- 1 mark for ratio of sides
- 2 marks for correct reasoning (including vertically opp angles)

(ii) All matching angles in similar triangles are equal $\therefore \angle ABP = \angle PCD$

These are alternate angles $\therefore AB \parallel CD$

- 1 mark for $\angle ABP = \angle PCD$ with correct reason
- 2 marks for alternate angles with reason

Question 14 (cont.)

c) $P = P_0 e^{kt}$

(i) $t=0$ $P = 15000$

$$15000 = P_0 e^{k \times 0} \therefore P_0 = 15000$$

$$P = 15000 e^{kt}$$

2010! $t=10$ $P=35000$

$$35000 = 15000 e^{10k}$$

$$e^{10k} = \frac{35}{15} \quad e^{10k} = \frac{7}{3}$$

$$10k = \ln \frac{7}{3}$$

$$k = \frac{1}{10} \ln \frac{7}{3}$$

$$k = 0.0847247 \dots$$

$$k \approx 8.5\%$$

- 1 mark for $P_0 = 15000$ or $P = 15000 e^{kt}$
- 2 marks for correct substitution $35000 = 15000 e^{10k}$
- 3 marks for $k = 0.08472 \dots$ or $k = 8.5\%$

(ii) $P = 50000$

$$P = 15000 e^{0.085t}$$

$$50000 = 15000 e^{0.085t}$$

$$e^{0.085t} = \frac{50}{15} = \frac{10}{3}$$

$$0.085t = \ln \frac{10}{3}$$

$$t = \frac{1}{0.085} \ln \frac{10}{3}$$

$$t = 14.16438 \dots$$

after 15 years

- 1 mark for $50000 = 15000 e^{0.085t}$
- 2 marks for $t = 15$ years or $t = 14.2$ years

Question 15

(15) a) $v = \frac{4}{2t+1} \text{ m/s}$

(i) initially: $t=0$ $x=-2$

$$x = \int \frac{4 dt}{2t+1} = \frac{4}{2} \int \frac{2 dt}{2t+1} = 2 \ln(2t+1) + C$$

$$t=0: \quad x = 2 \ln 1 + C = -2$$

$$C = -2$$

$$x = 2 \ln(2t+1) - 2$$

- 1 mark for correct integration + C
- 2 marks for correct value of C

(ii) $x=0$

$$2 \ln(2t+1) - 2 = 0$$

$$\ln(2t+1) = 1$$

$$2t+1 = e$$

$$2t = e - 1$$

$$t = \frac{e-1}{2} \text{ sec.}$$

- 1 mark for solving to $2t+1=e$
- 2 marks for correct value of t .

Question 15 (cont.)

$$(iii) \quad v = \frac{4}{2t+1} = 4(2t+1)^{-1}$$

$$a = -4(2t+1)^{-2} \times (2) \\ = -\frac{8}{(2t+1)^2}$$

$$(2t+1)^2 > 0 \quad -8 < 0 \\ \therefore \underline{\underline{a < 0}}$$

- 1 mark for correct differentiation
- 2 marks for correct reasoning to prove $a < 0$

b) (i) $P(1,4) = \frac{1}{3} \times \frac{3}{5} = \frac{1}{5}$

- 1 mark for correct answer

(ii) $P(\text{sum less than 5}) = P(1,1) + P(1,3) + P(3,1)$
 $= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{5} + \frac{1}{6} \times \frac{2}{5}$
 $= \frac{2}{15} + \frac{1}{15}$
 $= \frac{3}{15}$
 $= \frac{1}{5}$

- 1 mark for correct combinations and one correct prob.
- 2 marks for two other correct probabilities

Question 15 (cont.)

$$c) \quad P(12, 12) \quad x^2 = 12y$$

$$(i) \quad y = \frac{x^2}{12}$$

$$\frac{dy}{dx} = \frac{2x}{12} = \frac{x}{6}$$

$$\text{at } x=12 \quad m_t = \frac{12}{6} = 2$$

$$y - 12 = 2(x - 12)$$

$$y - 12 = 2x - 24$$

$$2x - y - 12 = 0$$

- 1 mark for gradient
- 2 marks for equation of the tangent

$$(ii) \quad 4a = 12 \quad \therefore a = 3$$

$$\text{direct. } y = -3$$

$$2x - y - 12 = 0$$

$$y = -3$$

$$2x + 3 - 12 = 0$$

$$2x = 9$$

$$x = \frac{9}{2}$$

$$D\left(\frac{9}{2}, -3\right)$$

- 1 mark for clearly showing the y-value of D
- 2 marks for x-value of D

Question 15 (cont.)

(iii) $\angle PFD = 90^\circ$?

$F(0, 3)$ $D(\frac{9}{2}, -3)$ $P(12, 12)$

$$m_{FD} = \frac{6}{-\frac{9}{2}} = -\frac{12}{9} = -\frac{4}{3}$$

$$m_{FP} = \frac{12-3}{12-0} = \frac{9}{12} = \frac{3}{4}$$

$$m_{FD} \times m_{FP} = -\frac{4}{3} \times \frac{3}{4} = -1$$

$\therefore FD \perp FP$

$\therefore \angle PFD = 90^\circ$

- 1 mark for gradients of FP and FD
- 2 marks for ^{clearly} showing $FP \perp FD$.

Question 16

$$a) (i) \quad 2\sqrt{x} = \frac{x}{2} \quad \therefore \quad 4x = \frac{x^2}{4} \quad / \times 4$$

$$x^2 - 16x = 0$$

$$x(x-16) = 0$$

$$x = 0, \quad x = 16$$

$$\text{for } x = 16 \quad y = \frac{16}{2} = 8$$

$$P(16, 8)$$

• 1 mark for CORRECT ANSWER

$$(ii) \quad V = \pi \int_0^{16} \left((2\sqrt{x})^2 - \left(\frac{x}{2}\right)^2 \right) dx$$

$$= \pi \int_0^{16} \left(4x - \frac{x^2}{4} \right) dx$$

$$= \pi \left[4 \frac{x^2}{2} - \frac{x^3}{12} \right]_0^{16}$$

$$= \pi \left[2x^2 - \frac{x^3}{12} \right]_0^{16}$$

$$= \pi \left[2 \times 16^2 - \frac{16^3}{12} \right] =$$

$$= \pi \left(512 - \frac{4096}{12} \right)$$

$$= \pi \left(512 - \frac{1024}{3} \right)$$

$$= \frac{512\pi}{3}$$

- 1 mark for CORRECT SIMPLIFIED EXPRESSION FOR THE VOLUME INTEGRAL
- 2 marks for CORRECT ANSWER

Question 16 (cont.)

b)

$$(1) \quad r = 12\% \text{ p.a.} = 0.12$$

$$r = 0.01 \text{ per month}$$

$$n = 5 \times 12 = 60$$

First annuity will earn the interest for 60 months

$$A_1 = 150(1+0.01)^{60} \\ = 150 \times 1.01^{60}$$

Second annuity will earn the interest for 59 months

$$A_2 = 150 \times 1.01^{59}$$

60th annuity will earn the interest for 1 month

$$A_{60} = 150 \times 1.01$$

$$\text{Total} = A_1 + A_2 + \dots + A_{60}$$

$$= 150 \times 1.01^{60} + 150 \times 1.01^{59} + \dots + 150 \times 1.01$$

$$= 150 (1.01^{60} + 1.01^{59} + \dots + 1.01)$$

$$= 150 (1.01 + 1.01^2 + \dots + 1.01^{60})$$

$1.01 + 1.01^2 + \dots + 1.01^{60}$ is a geometric series
with $a = 1.01$ $r = 1.01$

$$S_{60} = \frac{a(r^n - 1)}{r - 1} = \frac{1.01(1.01^{60} - 1)}{1.01 - 1}$$

$$T = 150 \times \frac{1.01(1.01^{60} - 1)}{1.01 - 1} = \$12372.95$$

Kate will save \$12372.95 after 5 years.

- 1 mark for CORRECT EXPRESSION FOR THE TOTAL INVESTMENT
- 2 marks for CORRECT ANSWER.

Question 1b (cont.)

(ii)

$$30000 = 150 \times \frac{1.01(1.01^n - 1)}{1.01 - 1}$$

$$1.01^n - 1 = \frac{30000(1.01 - 1)}{150 \times 1.01}$$

$$1.01^n = \frac{30000(1.01 - 1)}{150 \times 1.01} + 1$$

$$n = \log_{1.01} \left(\frac{30000(1.01 - 1)}{150 \times 1.01} + 1 \right)$$

$$n = \frac{\ln \left(\frac{30000(1.01 - 1)}{150 \times 1.01} + 1 \right)}{\ln 1.01}$$

$$n = 109.744 \dots$$

$$n = 110 \text{ months}$$

- 1 mark for CORRECT EXPRESSION FOR 1.01^n
- 2 marks for CORRECT ANSWER.

c) (i)

$$1 - 2\sin x = 0$$

$$2\sin x = 1$$

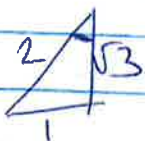
$$\sin x = \frac{1}{2}$$

$$x = 30^\circ, 180^\circ - 30^\circ$$

$$x = 30^\circ, 150^\circ$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}$$

$$0 \leq x < 2\pi$$

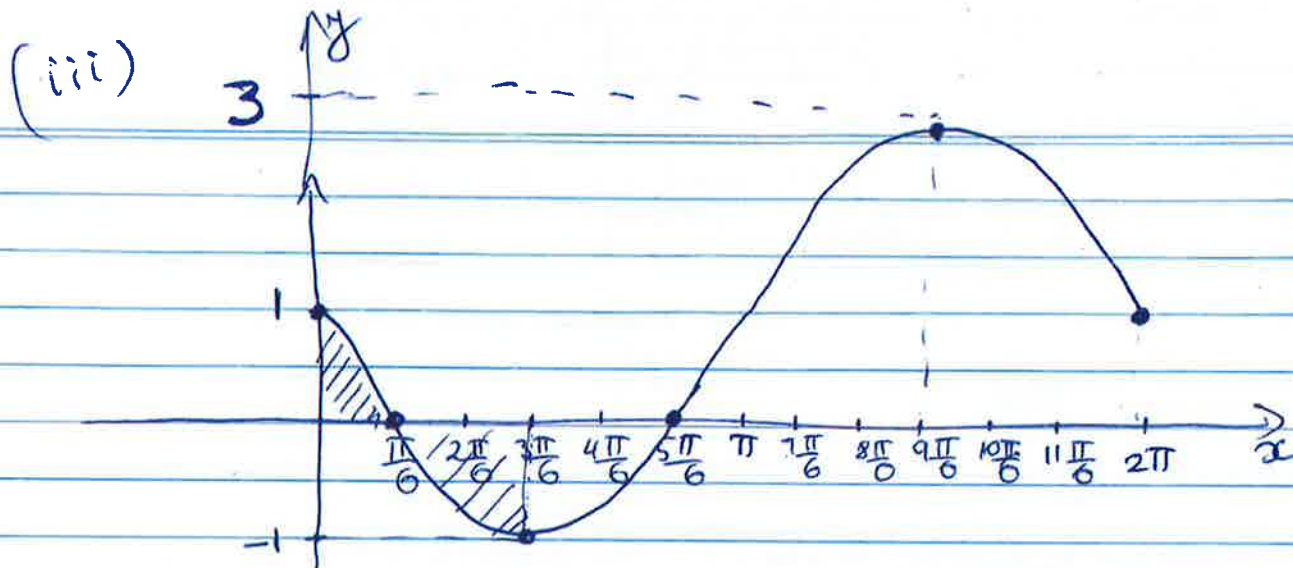


- 1 mark for CORRECT ACUTE SOLUTION $x = \frac{\pi}{6}$
- 2 marks for CORRECT ANSWER

Question 16 (cont.)

$$\begin{aligned} \text{(ii)} \quad x=0 \quad y &= 1 - 2\sin 0 = 1 \\ x=2\pi \quad y &= 1 - 2\sin 2\pi = 1 \end{aligned}$$

• 1 mark for CORRECT ANSWERS



$$-1 \leq \sin x \leq 1$$

$$-2 \leq 2\sin x \leq 2$$

$$-2 \leq 2\sin x \leq 2$$

$$-1 \leq 1 - 2\sin x \leq 3$$

- 1 mark for CORRECT SHAPE AND RANGE
- 2 marks for CORRECT CURVE, INCLUDING CORRECT INTERCEPTS AND ENDPOINTS.

Question 16 (cont.)

$$(iv) \quad A = \int_0^{\frac{\pi}{6}} (1 - 2\sin x) dx - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 - 2\sin x) dx$$



$$\begin{aligned} A &= \left[x + 2\cos x \right]_0^{\frac{\pi}{6}} - \left[x + 2\cos x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left[\frac{\pi}{6} + 2\cos\frac{\pi}{6} - 2\cos 0 \right] - \left[\frac{\pi}{2} + 2\cos\frac{\pi}{2} - \frac{\pi}{6} - 2\cos\frac{\pi}{6} \right] \\ &= \left[\frac{\pi}{6} + 2 \times \frac{\sqrt{3}}{2} - 2 \right] - \left[\frac{\pi}{2} - \frac{\pi}{6} - 2\frac{\sqrt{3}}{2} \right] \end{aligned}$$

$$\frac{\pi}{6} + \sqrt{3} - 2 - \frac{\pi}{2} + \frac{\pi}{6} + \sqrt{3}$$

$$\cancel{\frac{\pi}{6}} \frac{\pi}{3} - \frac{\pi}{2} + 2\sqrt{3} - 2$$

$$2\sqrt{3} - 2 - \frac{\pi}{6}$$

$$\approx 0.94 \text{ u}^2$$

- 1 mark for CORRECT INTEGRAL SPLIT INTO TWO AREAS BETWEEN $x=0$ TO $x=\frac{\pi}{6}$, AND $x=\frac{\pi}{6}$ TO $x=\frac{\pi}{2}$
- 2 marks for CORRECT INTEGRATION
- 3 marks for CORRECT ANSWER

- End of solutions -